

Image Recognition for Kids

How Do YOU Know This Is a Cat?



Humans See Cats Every Day

- We see cats at home, streets, and cartoons
- Our brain remembers what cats look like
- Just like a Tea seller knows regular customers!



Human vs Computer Vision

Human: Eyes see, brain thinks, learns from life.

Computer: Camera captures, program processes, learns from datasets.



What Is a Pixel?

Pixels are tiny dots of color. Every photo is made of thousands of these little colored squares!



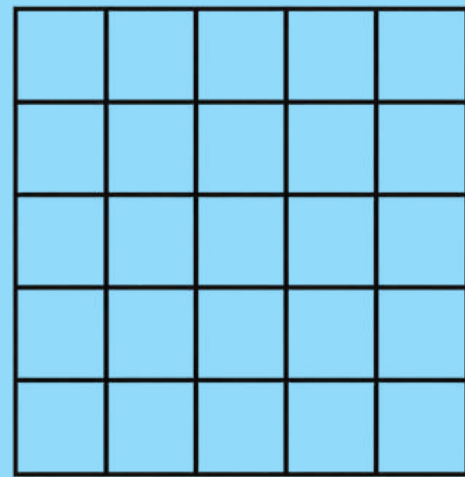
Pixels Are Numbers

Each pixel has a number! 0 means black, 255 means white. Colors use RGB - Red, Green, Blue - just like mixing Holi gulal powder!



Let's Make Pixel Art!

Draw your own 5x5 pixel cat or flower!
Color each square to create your
masterpiece. Start simple and watch your
art come alive!



Finding Patterns & Features

Computers look for special features to identify animals. Cats have pointy ears, whiskers, round pupils, soft fur, and a long tail!



Cat or Dog?

Like spotting a peacock by its crown and feathers, computers look for special features to tell animals apart!



Training the Computer

Computers learn by seeing 10 lakh labeled cat photos! Just like you practiced writing 'A' many times or learned cycling by trying again and again.



Getting Better Over Time

Computers make mistakes at first, but with practice, they get smarter! Just like you learning to write or ride a cycle.



Real Indian Applications of Image Recognition

Google Lens identifies flowers & text. Aadhaar Face ID verifies identity. Plantix helps farmers. Niramai detects health issues. Traffic cameras catch violations!



What We Learned!

1. Image recognition = computers identify pictures
2. Images are made of pixels (tiny colored squares)
3. Computers find patterns like ears and whiskers
4. Training uses thousands of example photos
5. Used in India: Google Lens, Aadhaar, Plantix!



Quick Quiz Time!

Q1: What are tiny dots in images? (Pixels!)

Q2: What do we call when computers study many photos? (Training!)

Q3: Name one Indian app using image recognition!

Shout your answers loud!





The next great AI inventor might be YOU!



Be the Next AI Inventor!

Homework: Use Google Lens to
identify 5 objects at home tonight!

